Available online at www.sciencedirect.com
SCIENCE DIRECT®
Cognitive Brain Research 23 (2005) 293 – 305
COGNITIVE BRAIN RESEARCH
www.elsevier.com/locate/cogbrainres

Research report

ERP analyses of task effects on semantic processing from words

Paloma Marí-Beffa[a,*], Berenice Valdés[a], Doug J.D. Cullen[a], Andrés Catena[b], George Houghton[a]

[a]School of Psychology, University of Wales, Bangor, Gwynedd LL57 2AS, United Kingdom
[b]University of Granada, Spain

Accepted 30 October 2004
Available online 23 December 2004

Abstract

Semantic (positive) priming refers to the facilitated processing of a probe word when preceded by a related prime word, and is a widely used technique for investigating semantic activation. However, the effect is interrupted or eliminated when attention is directed to low-level features of the prime word, such as its letters, a result which has been used to question the automaticity of semantic processing. We investigated this issue using both behavioural [reaction time (RT)] and electrophysiological measures [event-related potentials (ERPs)]. Subjects performed semantic categorization (living vs. nonliving) and letter search ("A" or "E") tasks on prime words followed by lexical decision on the probe. RT results showed the expected elimination of semantic priming following letter search. However, both prime tasks were affected by the semantic category of the prime, indicating that the meaning was processed. The ERP results supported this conclusion: an early component previously associated with automatic semantic processing [the Recognition Potential (RP)] was sensitive to the category of the prime word irrespective of the prime task. However, a later component (N400) was significantly affected by the task, in both the prime (categorization task) and probe words (semantic priming). The results dissociate rapid, automatic semantic processing from semantic priming. We suggest that a later inhibitory control mechanism suppresses this semantic activation when it is not relevant to the task, and that this produces the loss of semantic priming.

Theme: Neural basis of behavior
Topic: Cognition

Keywords: Automaticity; Semantic processing; Semantic priming; Category; ERPs; Task effects

1. Introduction

The impact of task set on the semantic processing of written words has been the focus of extensive debate in Cognitive Psychology and Neuropsychology [4,21,38,48,54,55]. In literate adults, understanding the meaning of a written word seems to be an automatic skill acquired through years of extensive practice. This view has received considerable experimental support. In the most well-known case, the automaticity of reading (compared to naming ink colours) is generally held to be the cause of the Stroop effect [36]. In addition, semantic processing apparently survives a number of other manipulations aimed at eliminating reading for meaning. In these studies, semantic priming is the most widely used technique. If processing a probe word is facilitated by the prior occurrence of a semantically related prime word, then the meaning of the prime is assumed to have been accessed. For example, semantic priming is still present even when the prime word is masked, so participants are unaware of its presence [37]. In addition, semantic priming is also evident when the prime word is presented some degrees of visual angle away from the focus of attention, in the parafovea, while participants are required to attend to an object located in the fovea [16]. Finally, semantic priming is also observed when participants are engaged in a concurrent task during the prime display [17]. Thus, it has been long

* Corresponding author. Fax: +44 1248 38 2599.
E-mail address: pbeffa@bangor.ac.uk (P. Marí-Beffa).

0926-6410/$ - see front matter © 2004 Elsevier B.V. All rights reserved.
doi:10.1016/j.cogbrainres.2004.10.016

believed that the semantic access from words can occur automatically without attention, or without explicit intention to engage in a semantic level task [49].

However, it has been found that if participants are asked not to read the word, but do something else with it, semantic priming from that particular prime word is reduced or eliminated [7,15,21,54,55] (i.e., this manipulation would not affect the semantic processing of a copresented distractor word in the prime display [38,39]). If any action can interrupt the semantic processing of a word, then this process would not be considered to be automatic. Thus, during the last two decades, most of the research in this field has tried to compare tasks that would enhance the access to the semantic properties of words (such as reading, semantic categorization or even lexical decision) with those that would hinder it by orienting the action away from this semantic content (searching for a letter in a word, counting the numbers of letters or case decision) [45,51]. More specifically, the reduction of semantic effects after searching the prime word for a letter is a fairly robust phenomenon that can be found with multiple procedures. It is important to note that, in most of these studies, semantic priming is the only measure of the processing of the meaning of a word. The main focus of the present study is to enlarge this view by comparing a number of measures of semantic processing following letter search.

1.1. Theoretical accounts of the impact of task on semantic processing

One of the first attempts to explain the impact of the prime task on the semantic processing of words came from the episodic memory literature, rather than from models of word reading. In the levels-of-processing theory, the recall of the prime word is enhanced if, at the time of encoding, the prime word received deep semantic processing. If the prime word were to be processed at a shallow, visual feature level, then its recall would be affected [12,13]. Smith et al. [54] used this framework to explain the reduction of semantic priming following a low-level task, analysing the semantic priming procedure as a memory problem of encoding and retrieval. Thus, from this point of view, the semantic processing of the prime word occurs during the prime display, when the word is encoded at a deep level. Although the specific mechanisms that enable this semantic analysis were not specified, this theory provided a useful framework where a number of prime task manipulations were tested. As a result, it was found that the shallow level of processing itself was not enough to produce the reduction in semantic priming. Rather, attention has to be directed to low-level intraword properties, for example, looking for a letter in a word. If participants are required to report the colour in which the prime word is written, then semantic priming survives [7], a result which accords with the usual interpretation of the Stroop effect. Given that the reduction in semantic priming reduction apparently involves only representations from the word domain, the next step was to study theoretical alternatives based specifically on models of word reading.

In such models, the semantic priming effect has been commonly explained as due to the spread of activation from the prime word to related words in a lexical net. When an associated probe word is presented, its higher level of activation facilitates a response [9]. From this perspective, semantic representations consist of this assembly of associated lexical nodes. Thus, the reduction or elimination of semantic priming involves the interruption (inhibition or decay) of the flow of activation across related lexical nodes [15,21,22]. It is important to highlight that, with an architecture of this nature, semantic processing could only be assessed by means of the semantic priming procedure. In this context, it is important to demonstrate that this semantic processing results from a status that is transferred from one word to a different one. In addition, the lack of representations of letters as entities differentiated from the lexical nodes makes it difficult to implement an interaction between letters and words that enables the reduction of semantic priming following letter search.

Stolz and Besner [55] used the McClelland and Rumelhart [46] hierarchical Interactive Activation (IA) model of word processing to explain the impact of the prime task on semantic priming effects. In Stolz and Besner's model, semantic access is achieved after orthographical (letter-based) and lexical levels are accessed. The lexical and semantic layers are reciprocally connected, so that activation of semantic nodes can spread back to the lexical level. For example, if the word CAT is presented, its lexical node would send activation to the superodinate category ANIMAL. Feedback to the lexical level would activate all the words connected to the category ANIMAL, for instance, DOG, thus producing semantic priming. Stoltz and Besner [55] propose that when attention is oriented to the letter level, the flow of activation to the semantic level is blocked. This blockage interrupts both access to the conceptual representation of the word and the flow of activation to associated words, reducing the semantic priming effect.

There is some controversy on the nature of this blocking mechanism. Marí-Beffa et al. [38,39] suggested that this mechanism is inhibitory in nature. From their point of view, the search for a letter within a word is not essentially different from the selection of a target (the letter) that is accompanied by distracting information (the word). Thus, in order to select the letter, the dominant task of reading the word needs to be controlled, using inhibition as this control mechanism [27,28]. From this point of view, both semantic access and spread of activation are automatic, but this initial activation and further propagation are then inhibited, reducing, eliminating, or even inverting the semantic priming effect. To support this, a number of studies have shown semantic negative priming following the search for a letter in a word [2,38,57].

In any case, other accounts are also compatible with Stoltz and Besner's IA architecture. In particular, the same outcome would be achieved if, instead of blocking or inhibiting access to semantics, the low-level task merely interrupted the feedback from the semantic level. Thus, semantic/conceptual processing of the prime word would still be automatic (i.e., not affected by the nature of the prime task), but the spread of activation to related nodes would be blocked or inhibited by the task. In the present study, we provide evidence that only the spread of activation to related words seems to be affected by the task, keeping intact the initial access to conceptual representations.

1.2. Category-specific effects: evaluating the semantic processing of a word while it is presented

One of the advantages of using a hierarchical architecture [10,46] is that it can predict that the letter search task can lead to semantic processing without semantic priming, by studying the conceptual activation of the prime word. To support this type of architecture, it is well known that words can produce category-specific effects. A number of Cognitive Neuropsychology studies have shown dissociations in the way the brain processes living and nonliving words and objects. For example, there are some patients who have problems when processing living objects [6,18].

When studying the undamaged brain, a number of neuroimaging studies have found different brain areas associated with living vs. nonliving objects [5,20,40,47,56]. There is some debate as to whether these areas respond to a localised representation specific for living objects, or whether they form part of a distributed network of brain systems activated by specific features of living objects (such as motor areas for motion). However, it seems clear that words for living objects may activate different brain systems following access to their meaning [20]. Finally, studies using event-related potentials (ERPs) suggest a possible dissociation between living and nonliving words. For example, Kiefer [31] found that words from natural categories showed an enhanced N400 in fronto-central and central–parietal electrode sites in a semantic priming task. Similarly, Weckerly and Kutas [59] also found an enhanced N400 and reduced P600 for animate sentences in frontal and central electrodes. These electrophysiological markers of semantic/conceptual processing can be used to investigate the impact of the level of the task on the semantic processing of words.

1.3. Electrophysiological correlates of semantic processing: automaticity reviewed

Event-related potentials (ERPs) have been widely used to study semantic processing in a variety of procedures. The most common index of semantic processing is the N400 component, a negative deflexion in the ERP wave, in a time window between 200 and 600 ms after stimulus onset, and with a centroparietal maximum [35]. This component has been associated to ongoing linguistic analysis such that, in general, every word elicits an N400, although the topography and latency of the N400 may change depending on the procedure. When used to study semantic priming from written words, the N400 component shows increased amplitude when an unrelated probe word is presented, compared to a related word. This is commonly addressed as the N400 effect [35]. A number of studies have been used to correlate N400 with controlled processes in semantic priming. For example, when manipulating the relatedness proportion in a lexical decision task, the N400 effect is greater with a high proportion of related trials compared to a low one [3,26]. In addition, the scalp distribution of the N400 seems to change depending on the structural aspects of semantic memory, so it can distinguish concrete and abstract words [33] and action verbs and nouns [14]. In the present study, we test whether the N400 effect is also sensitive to task manipulations when analysing both semantic priming and the differences between living and nonliving words.

The N400 component is not the only electrophysiological index of semantic processing. Indeed, many authors consider it too late to represent the automatic access to the meaning of a word. For example, it has been shown that the mapping from the orthographic representation of a word to its meaning can be achieved in less than 300 ms [30]. Recently, a number of studies have found semantic effects in a negative wave peaking around 250 ms after stimulus onset; this has been explicitly labelled as an index of word recognition, the Recognition Potential (RP; [41,53]). This component has been shown to be sensitive to several aspects of semantic processing, such as category [24,42,44], word difficulty [53], awareness [52], and concreteness [43]. More interestingly for our purposes, the RP does not seem to be affected by the level of processing of the task (semantic or shallow [22]), although it seems to change when the impact of this task is measured later in a subsequent word in a priming procedure [3].

In this study, we wanted to test the impact of the prime task on the semantic activation of the prime word and on the spread of activation to related probe words using both behavioural measures [reaction times (RTs) and accuracy] and event-related potentials. To do so, we focused on the electrophysiological response described previously as RP and N400 components. We implemented a prime-task methodology in which two displays are presented sequentially (Prime–Probe). In the prime display, two factors were manipulated, the category of the prime word (living or nonliving) and the type of task (letter search or categorization). To measure the task effects on semantic priming, a lexical decision task is performed on the probe display. Thus, semantic priming is measured by comparing responses to related and unrelated trials in this display as a function of the task performed on the primes.

If categorical semantic processing is automatic, then reaction times and brain responses to prime words may exhibit category-related differences irrespective of prime task. In addition, if semantic priming is due to automatic processing of the prime word, then we would expect to find behavioural and brain differences due to relatedness that do not change with the prime task. Conversely, nonautomatic processes involved in semantic processing would be identified through a significant interaction between task and any of the aforementioned semantic effects (category and relatedness).

2. Materials and methods

2.1. Participants

Twenty-six students of the University of Wales, Bangor (11 males and 15 females) aged from 18 to 30 years, took part in this experiment, obtaining course credits for their participation. All subjects were right-handed, with normal or corrected-to-normal vision, and were native English speakers with no reported reading difficulties.

2.2. Materials

The experimental tasks and EEG recording were conducted inside an acoustically shielded room. The experimenter monitored the experimental sessions from a separate room via two-way intercom and video cameras. Stimuli were displayed to the subject in a flat 17-in. digital monitor located inside the shielded room. A standard computer keyboard was used to collect the responses. Stimuli and behavioural responses were designed, controlled and recorded by E-prime software running on an IBM-compatible PC. This system also sent "trigger" signals to the EEG recording software (see below) synchronised with stimulus and response events.

2.3. EEG recording

EEGs were collected from 16 electrode sites of the International 10/20 System (F3, F4, C3, C4, P3, P4, O1, O2, PO7, PO8, FC5, FC6, CP5, CP6, FZ, PZ) referred to the vertex electrode (Cz). An elastic electrode-cap (Easy cap) with sintered-silver chloride electrodes was used for this purpose. A bipolar eye channel with two individual electrodes placed 1 cm above and below the left eye was used to detect vertical eye movements (VEOG channel). The electrodes output-signal was amplified with a 16-bit amplifier (Synamp) with a band-pass filter from 0.15 to 100 Hz and a gain of 500 times at a sampling rate of 1000 Hz. All electrode impedances were below 5 kΩ. The continuous EEG was recorded using Neuroscan Acquire software on an IBM-compatible PC. An interface with the PC controlling stimulus presentation allowed the time of task events and responses to be included in the EEG record.

2.4. Stimuli

The purpose of this experiment required pairs of related words subject to a number of constraints. The words were of high frequency, and the first member of each pair had to be classifiable as either living or nonliving (for the categorization task). In addition, the first member of each pair had to contain either the letter A or E (search target) but never both. The second word of the pair could not contain this letter. Note that in the letter search task, a target letter is always present, but it is necessary to determine whether it is A or E. By virtue of these constraints, exactly the same pairs of stimuli could be used for the categorization and for the letter search task.

The stimulus set consisted of 40 English words, 3 to 8 characters in length, extracted from the MRC Psycholinguistic Database [60]. Half the words were classified as living, and the other half as nonliving. Of the words in each of these semantic groups, half (10) contained the letter A (but not E) and half the letter E (but not A). A one-way analysis of variance (ANOVA) over the frequency of use [34] (F<1 for prime and probe displays) and association strength values of living and nonliving words (F<1 for both types of trials) revealed no significant differences between the word groups (see Appendix A).

Each of these 40 words was then paired with the first associatively related word found in the Associative Thesaurus of English database [32] that did not contain the prime search letter. The associative values from this database represent the percentages of first responded words in a free association task [32]. These were the Related pairs. The Unrelated pairs were created by re-pairing the prime and the probe words such that the new pairs shared no relationship. For the lexical decision task in the probe, we needed pseudowords, and 40 such items were extracted from the Lexicon Project Database [1]. The pseudowords were 4 to 8 characters long and with a mean of 3 orthographic neighbours in English (0 to 5). Pseudowords conditions were created only for control purposes and were not analysed.

The resulting 40 pairs of related words were then used to construct the lists of stimuli arranged in 4 factorial conditions, crossing Relatedness (Related/Unrelated) with Lexicality (word/pseudoword) with 40 trials per condition giving a total of 160 trials. Each prime word was presented four times (once in each condition) and the probe words were presented twice. Each pseudoword was also repeated twice in the probe displays. The whole set of stimulus pairs was presented twice in random order resulting in a total of 320 trials per experimental session. Clearly, the Related Pseudoword condition is meaningless, but the slot was filled with the equivalent Unrelated Pseudoword trials to balance the proportion of words and pseudowords in the design.

Thus, for the probe display, 50% of the stimuli were words and 50% were pseudowords.

2.5. Procedure

The prime tasks (Letter Search, Categorization) were manipulated between blocks. This is an important feature of the design as both tasks were performed on the same set of words. Mixing tasks within blocks on the same stimulus set gives rise to cross-talk interference [58], clearly something we need to avoid. The order of blocks was counterbalanced across participants, half performing the letter search block first, and the other half the categorization block. In order to analyse possible sequence effects, a between-groups factor (order of the tasks) was included in the analysis.

Instructions were displayed on the monitor of the computer at the beginning of the experiment, and were followed by 48 practice trials. Participants were instructed to use the left hand to respond to the prime task and the right hand to respond to the probe task. For the Letter Search prime task, pressing "C" with the left index finger indicated the presence of the letter E, and key "D" with the left middle finger the presence of the letter A. The same responses were used for the Categorization task, with "C" assigned to living words and "D" to nonliving. For the Lexical Decision probe task, in all cases pressing "M" with the right index finger indicated a real word, while pressing "K" with the right middle finger indicated a pseudoword.

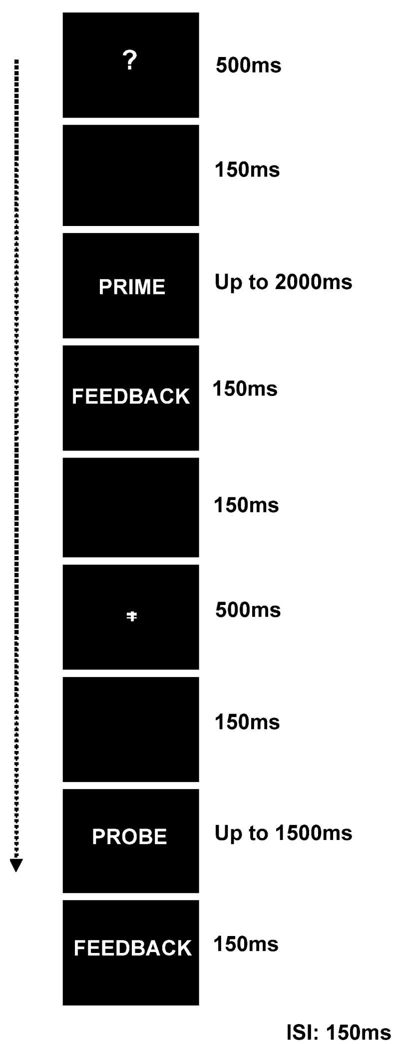

Fig. 1. Sequence of events in each trial. Fixations acted as cues to indicate a Letter Search or Categorization task in the prime display (?) and a Lexical Decision in the probe display (*).

The structure of the experimental trials is shown in Fig. 1. Each trial started with a question mark (?) at fixation for 500 ms. The prime display appeared after a blank interval of 150 ms and remained on the screen until the subject responded. This response had a maximum tolerance window of 2000 ms. This was followed by a blank interval of 150 ms, after which an asterisk (*) appeared at fixation for 500 ms. After a further blank interval of 150 ms, the probe stimulus was displayed and kept on the screen until response with a response window of 1500 ms. After each response, a 150-ms feedback display was presented containing a beep for incorrect responses or a blank screen for correct responses. The beginning of the next trial followed after a 150 ms Interstimulus Interval (ISI).

The total duration of the experimental task (two blocks of 160 trials) was approximately 45 min with time for a break between the blocks. Due to the technical requirements of preparing for the EEG recording, each experimental session lasted up to 2 h.

3. Results

3.1. Behavioural analysis

Participants committing more than 25% errors in any condition or with a performance difference between tasks higher than 25% were excluded from the analysis. As a result, a total of 22 participants remained in the analysis. Trials with reaction times faster than 250 ms were excluded and considered as incorrect responses.

3.2. Responses to prime displays: task and category effects

To test any possible influence of the sequence of the two tasks, a 2×2×2 (Sequence×Task×Category) ANOVA was performed with mean correct reaction times and the percentage of errors. Results showed no interaction of the factor Sequence with the Task×Category interaction neither in reaction times, $F(1,20)=1.588$, $p=0.22$, nor in the percentage of errors, $F(1,20)=1.67$, $p=0.21$.

A 2×2 (Task×Category) ANOVA of mean correct reaction times showed differences between responses to living and nonliving words, $F(1,20)=15.17$, $p<0.01$, that changed depending on the task, $F(1,21)=79.91$, $p<0.01$. Reaction time to living words was significantly faster than for nonliving words in Categorization task, $F(1,21)=66.18$,

Table 1
Mean RTs, percentage of errors and standard deviations (S.D.) for prime responses

	Categorization			Letter search		
	Living	Nonliving	NL−L	Living	Nonliving	NL−L
Mean	769.53	890.80	121.27*	901.55	861.31	−40.24*
% Errors	11.14	10.00	−1.14	11.59	8.52	−3.07
S.D.	132	168		202	187	

* $p<0.05$.

$p<0.01$, but the opposite pattern was found for the Letter Search task, $F(1,21)=10.29$, $p<0.01$, where RT to living words was significantly slower than to nonliving words (see Table 1). An additional Task×Relatedness ANOVA conducted for the percentage of errors in the prime showed no significant effects.

3.3. Responses to probe displays: semantic priming and the prime task effect

For the analysis of the probe responses, we collapsed trials with living and nonliving prime words. This was done after verifying that the category of the prime word did not affect the probe response and did not interact with any other variable in the design.

Mean correct RT and percentage of errors per participant and condition were submitted to two separate 2×2×2 (Sequence×Task×Relatedness) ANOVAs to test any possible effects of sequence. Results showed no significant interaction of the sequence factor with the Task×Related-Relatedness interaction either with the RTs, $F(1,20)=0.406$, $p=0.53$, or with the errors, $F(1,20)=0.38$, $p=0.55$.

The 2×2 (Task×Relatedness) ANOVA of mean correct RT showed that the semantic priming effect was globally nonsignificant, $F(1,21)=3.94$, $p=0.60$, although it significantly changed depending on the task, $F(1,21)=5.51$, $p<0.05$. There were 32 ms of reliable semantic priming effect when the prime task was Categorization, $F(1,21)=6.66$, $p<0.05$, but no such effect (−2 ms) when a letter search was performed on the prime display, $F(1,21)=0.065$, $p=0.80$ (see Table 2). An identical two-way (Task×Relatedness) ANOVA was conducted for the percentage of errors showing no significant effects of either factor or their interaction.

To summarize, the behavioural data showed the standard reduction of semantic priming when participants searched the prime word for a letter. In addition, the data showed strong differences between living and nonliving words. Living words seemed to be more salient than the nonliving ones, leading to faster responses when categorizing them, and slower responses when being searched for a letter. While the Letter Search task eliminated semantic priming in the probe, it did not remove differences between living and nonliving words in the primes. Instead, these differences were inverted, compared to the Categorization task. Hence, letter search in a word is sensitive to the semantic category of the word. These behavioural results alone suggest a strong dissociation between semantic activation and spread of activation to associated words (priming). This dissociation was further explored with the analysis of the psychophysiological responses.

Table 2
Mean RTs, percentage of errors and standard deviations (S.D.) for probe responses (Lexical Decisions)

	Categorization			Letter search		
	R	UR	UR−R	R	UR	UR−R
Mean	819.49	851.51	32.02*	834.43	832.37	−2.06
% Errors	9.89	11.25	1.36	10.23	9.89	−0.34
S.D.	186	207		165	160	

R=Related, UR=Unrelated.
* $p<0.05$.

3.4. Event-related potentials (ERP) analysis

Continuous EEG recordings were filtered off-line with a band-pass filter from 1 to 30 Hz. EEG epochs of 1600 ms, time-locked to stimulus onset, were extracted for correct responses in both prime and probe stimuli separately. In both cases, an 800-ms prestimulus interval was used to include the prefixation interval for the baseline correction[1]. Eye-blink artefacts were corrected from the raw data using a linear derivation transform. Baseline correction was applied for the 150-ms prefixation interval (−800 to −650 ms) and all epochs were then re-referenced to an average reference.

For each participant, all trials containing wrong responses or activity ±50 μV were excluded from the averaging within condition. Each average contained a minimum of 25 valid trials per participant.

Visual inspection of the ERP waves for both prime and probe words showed a large positive component at around 100 ms and a large negative one around 200 ms over posterior regions. Smaller components can be also observed around 300–400 ms distributed mainly in centroparietal regions.

[1] Due to the short interval between the cue and the stimulus, we decided to use the precue interval as baseline. This was done to include preparatory effects that start with the onset of the cue (the search letter or the asterisk) but that are also part of the processing of the word itself. This procedure has been already used in studies where preparatory effects can affect the processing of a particular stimulus [19,29].

A criterion of 15 ms consecutive significant results (p<0.025) was used for the report.

3.5. Responses to prime displays

For each electrode and at each millisecond, a 2×2 (Task×Category) ANOVA was performed over the voltage values of the stimulus interval (−100 to 800 ms). According to this criterion, statistical analysis showed significant effects at two different latencies. Living and nonliving words showed differences in amplitude at around 200 ms both in letter search and categorization trials. Nonliving words generated larger amplitudes than living words in frontal and posterior regions. Although the letter search trials apparently showed greater differences than the categorization trials, this interaction was not significant (see Fig. 2).

A second group of differences can be observed around 400–500 ms. Categorization trials showed differences between living and nonliving words over an extended area in parietal and right frontal areas with greater amplitudes for living words. This difference was not observed in Letter Search trials (See Fig. 3). The interaction between task and category was significant at around 500 ms.

From this pattern of results, it is possible to suggest a distinction between early automatic semantic effects, present in both Letter Search and Categorization trials (at around 200 ms), and later, more controlled semantic effects that were present only in the Categorization trials (at around 400–500 ms).

3.6. Responses to probe displays

Following the analysis procedure of the behavioural data, we collapsed trials with living and nonliving prime words. We tested first that the category of the prime words did not have an effect on the probe responses nor interacted with other variables in the design.

For each electrode and at each millisecond, a 2×2 (Task×Relatedness) ANOVA was performed over the voltage values of the stimulus interval (0 to 800 ms). The inspection of the ANOVA results showed differences between related and unrelated trials from 300 to 600 ms mainly in central and parietal regions. In agreement with previous findings on standard semantic priming, the Categorization trials showed larger negative amplitudes for unrelated than for related words. This effect was significant only over parietal regions from around 300 ms up to around 600 ms. This pattern was inverted in the Letter Search trials where unrelated words generated less negative amplitudes than the related ones over the same sites and within the same time window. This change in the direction of semantic priming depending on the task was significant in parietal areas at around 500 ms (see Fig. 4).

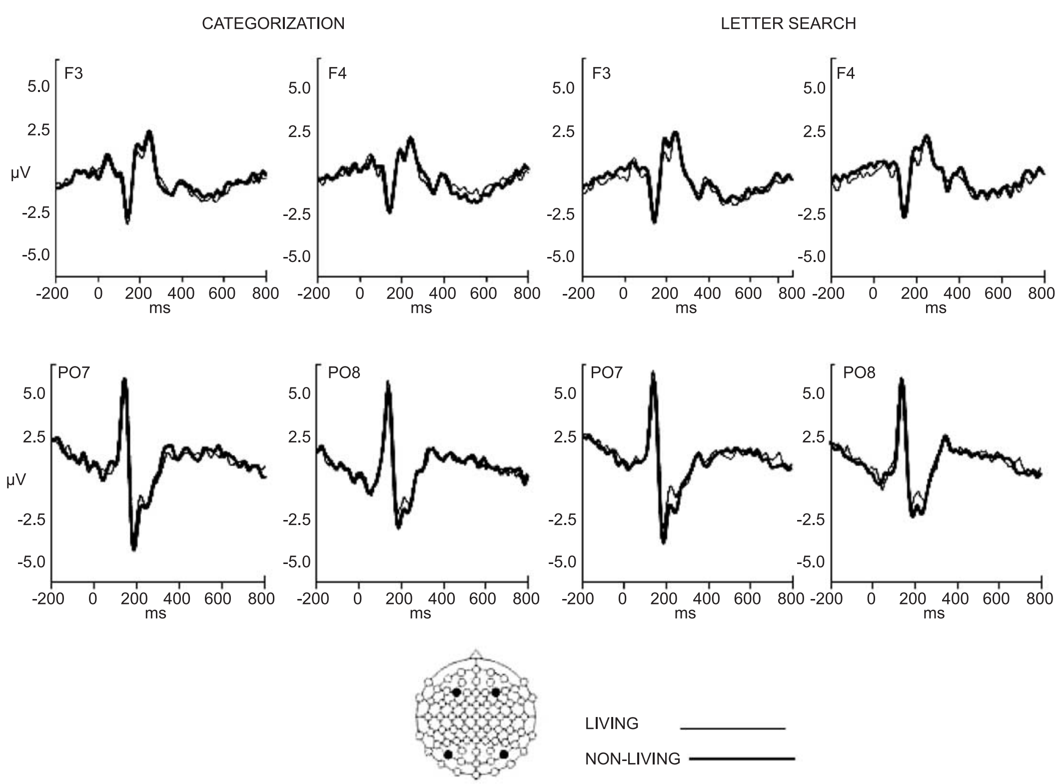

Fig. 2. ERP waveforms recorded during prime-task interval showing the category effects (Living, thin line, vs. Nonliving, bold line) in the Categorization and Letter Search blocks. The figure also shows the electrode positions where the category differences were greater at around 200 ms.

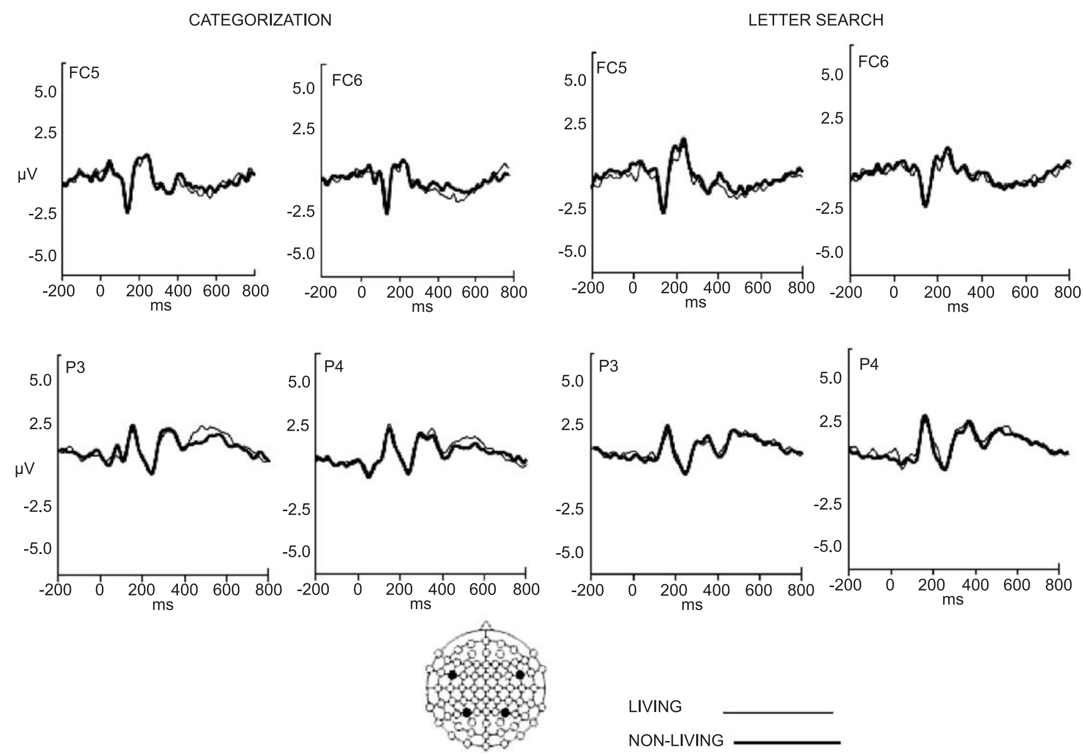

Fig. 3. ERP waveforms recorded during prime-task interval showing the category effects (Living, thin line, vs. Nonliving, bold line) in the Categorization and Letter Search blocks. The figure also shows the electrode positions where the category differences were greater at around 500 ms.

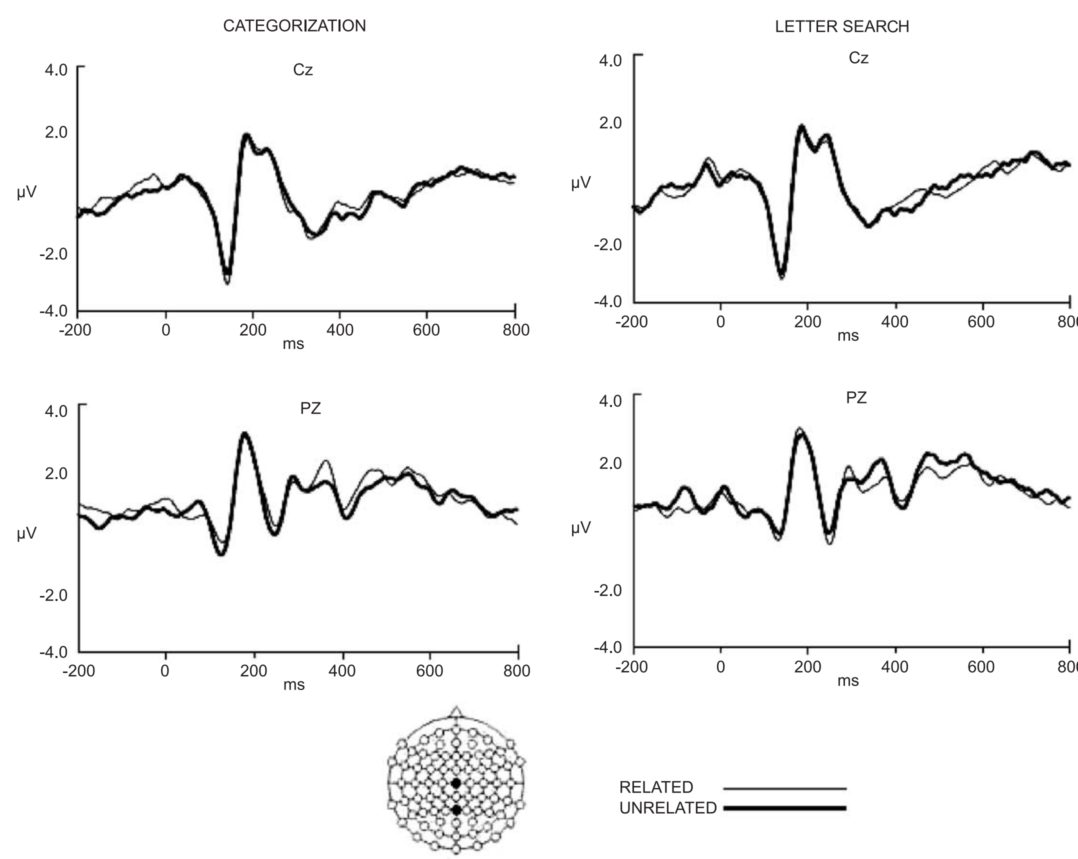

Fig. 4. ERP waveforms recorded during the probe interval (Lexical Decisions) showing semantic priming effects (Related in thin line and Unrelated in bold line) over central Cz and parietal PZ regions in both Categorization and Letter Search blocks.

4. Discussion

The main purpose of the present study was to investigate the role of task goal on word processing. In particular, we sought converging evidence that the automatic semantic processing of words is dissociable from such "knock-on" effects as semantic priming (and hence that the absence of such an effect is not good evidence of an absence of semantic processing of prime words [38,39]). To provide converging measures, we employed both behavioural and ERP analyses. We discuss the results regarding the prime and probe words separately.

4.1. The effect of task on semantic processing of the prime words

Despite the behavioural replication of the elimination of semantic priming with letter search, the processing of the prime word was sensitive to its semantic category (living vs. nonliving) whatever the task. In particular, letter search in a word is affected by the semantic category of the word, indicating that letter search does not eliminate semantic activation, although it does not lead to priming.

Interestingly, the precise effect of semantic category depended on the prime task: when participants had to categorize the prime words they were faster on living words, but when they had to identify a letter in the same words, they were impaired. The precise reason why the effect should go in opposite directions is not clear on the basis of these present results alone. Living and nonliving objects may differ in a number of aspects, such as emotional impact (e.g., MOTHER vs. TABLE). In addition, living words may have a lower concreteness level than nonliving words. The important point in the present context is that the differences between these two categories can only be explained as a result of some aspect of semantic processing, be it automatic (in letter search) or controlled (in categorization).

While living words facilitate a semantic level task, such as categorization, they act as an increased source of distraction when attention needs to be directed away from the word level to the letter level. One possible explanation that we would like to discount is that while performing letter search, participants were distracted by having performed a categorization task (the easier task) on the same stimuli. There is much evidence that performing different tasks on the same stimulus set can lead to cross-talk, especially when the two tasks are mixed within-block in a switching design [58]. For this reason, we deliberately separated the two prime tasks by block with a rest period in between. In addition, task order (counterbalanced across participants) was included in the analysis and did not interact with the factors of interest. The present data thus provide no support for an explanation based on cross-task interference.

The conclusion drawn from the behavioural data, that the meaning of the prime words was accessed irrespective of the task, is supported by the ERP data. Analysis of the ERPs generated by the prime words showed differences between the semantic categories in the Recognition Potential (RP), an ERP component occurring around 200 ms after stimulus onset. As discussed in the Introduction, previous work has shown this component to be related to word recognition and semantic processing [41]. Most importantly, the Categorization and Letter search tasks produced the same effect on this component of the ERP, supporting the view that this initial semantic access is automatic, occurring irrespective of what task is performed on the words [25]. As mentioned before, living and nonliving words may differ in a number of aspects, affecting the amplitude of this component. For example, previous studies have shown that the amplitude of the RP is greater for concrete words than for abstract words [43]. We compared our living and nonliving words in their concreteness ratings [11] and, although the difference was not significant (p=0.6), living words had slightly lower concreteness ratings than nonliving words. In any case, the RP differences between these two categories were not affected by the task, suggesting the automatic processing of these semantic aspects.

However, later ERP components show a clear dissociation between the two tasks with respect to the semantic category of the stimulus. When participants are required to allocate their attention to a word's meaning (in the Categorization task), differences between living and nonliving words appear in a slow positive component with maximum peak around 300 ms centred over parieto-central electrodes. Living words are more positive than nonliving words. We propose that this difference reflects top–down mechanisms guided by the task goals. The larger positivity for living words may reflect a strategic mechanism, in that participants perform the task by looking for living words, essentially representing the living category as a "yes" response. This suggestion fits with the significant RT advantage enjoyed by living words. The important point though is that when participants are categorizing the stimuli, effects due to differences in semantic category can be found in both later and earlier ERP components, reflecting the elaborated semantic processing necessary to convert the activation of a word's meaning into a conscious judgement about what category it belongs to. In the case of the letter-search task, no semantic category differences were found beyond the RP difference discussed above. We propose that this is because the word's meaning is irrelevant to the task and hence nonautomatic semantic processing does not occur.

4.2. The effect of the prime task on the probe

As noted above, the behavioural results replicate the standard prime task effect, whereby semantic priming is eliminated when a Letter Search is performed on the prime word. Also in agreement with previous work, in the ERP data, semantic priming is characterized by greater amplitudes in a negative wave peaking around 400 ms after probe onset (N400). This N400 semantic priming effect can be clearly observed following a Categorization prime. In contrast, following Letter Search, the pattern is inverted and unrelated words tend to produce a more positive wave than related ones. These differences demonstrate that semantic priming does take place in both the categorization and letter search tasks, but the direction of the differences is inverted by the task. This result is compatible with previous theories suggesting that, in the letter search task, inhibitory control is exerted upon the already processed semantic representations of the prime word in order to attend to the letter level [38,39,57]. As in previous work [8,23], the shape and magnitude of the N400 seems to be affected by an overlapping of this wave with a long positive complex (LPC) which is observed between 400 and 600 ms (see Fig. 4).

It was surprising to find that the RP component was not sensitive to the priming manipulation; its amplitude being the same for both related and unrelated trials across the two tasks. Apparently, priming effects are reflected at later stages of processing than are categorical effects. One possibility is that priming procedures have an influence in later stages of response selection reflected by the N400 component. This interpretation is in agreement with some previous work that attributes semantic priming effects in lexical decision tasks to more strategical processes [50]. Another possibility is that the RP is only sensitive to certain aspects of categorical semantic processing that are evident, for example, when comparing living and nonliving words, or concrete vs. abstract words. The probe words were a combination of living and nonliving words; therefore, no RP effects would be expected.

To sum up, the *later* differences in the prime ERP produced by the Categorization task, and dependent upon semantic category, are uniquely associated with the standard ERP correlates of behavioural semantic priming in the probe. Hence, we conclude that finding semantic priming in this type of design is dependent upon the kind of elaborated semantic processing required for making a categorization decision. When such a decision is not needed, this activation does not take place, or is actively suppressed.

5. Conclusions

In sum, our results, both with reaction times and ERP help to dissociate an early stage of semantic processing which occurs automatically and correlates with the Recognition Potential (N200) and a later stage where activation is controlled by the task goals (N400). Overall, we observed early (200 ms) categorical effects that do not seem to depend on task goals and late category and semantic priming effects (400–500 ms) that change size or direction depending on the task. This pattern of results supports the idea that conceptual activation and spread of activation are relatively independent processes. Only the first one would be completely unaffected by the letter search task (at least at 200 ms), while this task reverses the effect for the second one (measured through semantic priming). Following this pattern of results, it is possible to suggest that the top–down control over the flow of semantic information does not have a strong impact of bottom–up semantic processing (or forward connections, following McClelland and Rumelhart [46], IA model). Instead, this control seems to be exerted on feedback pathways, possibly inhibitory in nature, that are used to modulate the activation of different types of representations for a particular response.

Acknowledgments

This research was supported by grant No. 5/S16740 from Biotechnology and Biological Sciences Research Council (BBSRC) to Dr. Paloma Marí-Beffa and by the BBSRC Underwood Fund to Dr. Andrés Catena and Dr. Paloma Marí-Beffa. We thank Karen Howard for helping during the acquisition of the data.

Ethics Statement: This study has been performed in accordance with the ethical standards laid down in the 1964 Declaration of Helsinki. All participants gave their informed consent prior to their inclusion in the study.

Appendix A

Letter	Prime	Prime frequency	Probe	Probe frequency	% Association
Nonliving words					
E	VEHICLE	35	CAR	278	82
E	KNIFE	76	FORK	15	45
E	STREET	244	ROAD	212	35
E	BIBLE	59	BOOK	199	35
E	CENT	158	DOLLAR	58	32
E	CEILING	31	FLOOR	163	29
E	TEXTILE	28	CLOTH	46	23
E	BLUE	143	SKY	62	20
E	HOTEL	126	ROOM	398	13
E	TIE	25	KNOT	10	25

Appendix A (*continued*)

Letter	Prime	Prime frequency	Probe	Probe frequency	% Association
Nonliving words					
A	LAMP	18	LIGHT	351	48
A	BOAT	72	SHIP	84	17
A	SUGAR	34	SWEET	81	44
A	SALT	46	PEPPER	13	43
A	JAIL	21	BIRD	32	24
A	MAIL	47	POST	117	20
A	HOSPITAL	110	BED	135	20
A	FLAG	16	POLE	18	16
A	BAR	85	DRINK	84	32
A	LAW	299	ORDER	379	12
		83.65		**136.75**	**30.75**
Living words					
E	SISTER	38	NUN	2	62
E	SINGER	10	SONG	72	38
E	ENEMY	88	WAR	492	34
E	COUPLE	122	TWO	1510	31
E	PRINCESS	10	CROWN	20	23
E	NURSE	17	DOCTOR	100	18
E	PEOPLE	847	CROWD	53	16
E	PRIEST	16	CHURCH	350	14
E	CLIENT	15	SOLICITOR	6	5
E	QUEEN	41	KING	90	40
A	HUSBAND	131	WIFE	231	85
A	AUNT	22	UNCLE	57	61
A	BUFFALO	16	BILL	143	45
A	POPULATION	136	EXPLOSION	16	32
A	RAT	9	MOUSE	10	28
A	RAM	2	SHEEP	24	24
A	MILITANT	8	STUDENT	136	17
A	PLANT	125	FLOWER	24	16
A	ANIMAL	69	DOG	77	12
A	FAMILY	331	HOME	566	14
		102.65		**198.95**	**30.75**

References

[1] D.A. Balota, M.I. Cortese, K.A. Hutchison, B. Loftis, J.H. Neely, D. Nelson, G.B. Simpson, R. Treiman, The English Lexicon Project: A Web-Based Repository of Descriptive and Behavioral Measures for 40,481 English Words and Nonwords, Washington University, 2002, http://elexicon.wustl.edu/.

[2] D. Besner, M.C. Smith, C.M. MacLeod, Visual word recognition: a dissociation of lexical and semantic processing, Journal of Experimental Psychology. Learning, Memory, and Cognition 16 (5) (1990) 862–869.

[3] C.M. Brown, P. Hagoort, D.J. Chwilla, An event-related brain potential analysis of visual word priming effects, Brain and Language 72 (2) (2000) 158–190.

[4] C.M. Bushell, Dissociated identity and semantic priming in Broca's aphasia: how controlled processing produces inhibitory semantic priming, Brain and Language 55 (2) (1996) 264–288.

[5] S.F. Cappa, D. Perani, T. Schnur, M. Tettamanti, F. Fazio, The effects of semantic category and knowledge type on lexical–semantic access: a PET study, Neuroimage 8 (4) (1998) 350–359.

[6] A. Caramazza, J.R. Shelton, Domain-specific knowledge systems in the brain: the animate–inanimate distinction, Journal of Cognitive Neuroscience 10 (1998) 1–34.

[7] P.R. Chiappe, M.C. Smith, D. Besner, Semantic priming in visual word recognition: activation blocking and domains of processing, Psychonomic Bulletin and Review 3 (4) (1996) 249–253.

[8] D.J. Chwilla, P. Hagoort, C.M. Brown, The mechanism underlying backward priming in a lexical decision task: spreading activation versus semantic matching, Quarterly Journal of Experimental Psychology: Human Experimental Psychology 51 (3) (1998) 531–560.

[9] A.M. Collins, E.F. Loftus, A spreading-activation theory of semantic processing, Psychological Review 82 (1975) 407–428.

[10] A.M. Collins, M.R. Quillian, Retrieval time from semantic memory, Journal of Verbal Learning and Verbal Behavior 8 (2) (1969) 240–247.

[11] M. Coltheart, The MRC psycholinguistic database, Quarterly Journal of Experimental Psychology 33A (1981) 497–505.

[12] F.I. Craik, R.S. Lockhart, Levels of processing: a framework for memory research, Journal of Verbal Learning and Verbal Behavior 11 (6) (1972) 671–684.

[13] F.I. Craik, E. Tulving, Depth of processing and the retention of words in episodic memory, Journal of Experimental Psychology. General 104 (3) (1975) 268–294.

[14] S. Dehaene, Electrophysiological evidence for category-specific word-processing in the normal human brain, NeuroReport 6 (16) (1995) 2153–2157.

[15] F.J. Friedrich, A. Henik, J. Tzelgov, Automatic processes in lexical access and spreading activation, Journal of Experimental Psychology: Human Perception and Performance 17 (1991) 792–806.

[16] L.J. Fuentes, P. Tudela, Semantic processing of foveally and parafoveally presented words in a lexical decision task, Quarterly Journal of Experimental Psychology 45A (1992) 299–322.

[17] L.J. Fuentes, E. Carmona, I.F. Agis, A. Catena, The role of anterior attention system in semantic processing of both foveal and parafoveal words, Journal of Cognitive Neuroscience 6 (1994) 17–25.

[18] G. Gainotti, M.C. Silveri, Cognitive and anatomical locus of lesion in a patient with a category-specific semantic impairment for living beings, Cognitive Neuropsychology 13 (1996) 357–389.

[19] C.M. Gómez, J. Marcos, C. Grau, Preparatory visuo-motor cortical network of the contingent negative variation estimated by current density, Neuroimage 20 (2003) 216–224.

[20] O. Hauk, F. Pulvermuller, Effects of word length and frequency on the human event-related potential, Clinical Neurophysiology 115 (5) (2004) 1090–1103.

[21] A. Henik, F.J. Friedrich, W.A. Kellogg, The dependence of semantic relatedness effect upon prime processing, Memory and Cognition 11 (1983) 366–373.

[22] A. Henik, F.J. Friedrich, J. Tzelgov, S. Tramer, Capacity demands of automatic processes in semantic priming, Memory and Cognition 22 (22) (1994) 157–168.

[23] H. Hill, M. Strube, D. Roesch-Ely, M. Weisbrod, Automatic vs. controlled processes in semantic priming: differentiation by event-related potentials, International Journal of Psychophysiology 44 (3) (2002) 197–218.

[24] J.A. Hinojosa, M. Martín-Loeches, F. Muñoz, P. Casado, C. Fernández-Frías, M.A. Pozo, Electrophysiological evidence of a semantic system commonly accesses by animals and tools categories, Cognitive Brain Research 12 (2001) 321–328.

[25] J.A. Hinojosa, M. Martín-Loeches, F. Muñoz, P. Casado, M.A. Pozo, Electrophysiological evidence of automatic early semantic processing, Brain and Language 88 (2004) 39–46.

[26] P.J. Holcomb, Automatic and attentional processing: an event-related brain potential analysis of semantic priming, Brain and Language 35 (1) (1988) 66–85.

[27] G. Houghton, S.P. Tipper, A model of inhibitory mechanisms in selective attention, in: D. Dagenbach, T. Carr (Eds.), Inhibitory Processes in Attention, Memory and Language, Academic Press, San Diego, CA, 1994, pp. 53–112.

[28] G. Houghton, S.P. Tipper, Inhibitory mechanisms of neural and cognitive control: applications to selective attention and sequential action, Brain and Cognition 30 (1996) 20–43.

[29] S. Hsieh, Y. Yu, Switching between simple response-sets: inferences from the lateralized readiness potential, Cognitive Brain Research 17 (2) (2003) 228–237.

[30] M.A. Just, P.A. Carpenter, The Psychology of Reading and Language Comprenhension, Allyn and Bacon, Boston, 1987.

[31] M. Kiefer, Perceptual and semantic sources of category-specific effects: event-related potentials during picture and word categorization, Memory and Cognition 29 (1) (2001) 100–116.

[32] G.R. Kiss, C. Armstrong, R. Milroy, J. Piper, An associative thesaurus of English and its computer analysis, in: A.J. Aitkin, R.W. Bailey, N. Hamilton-Smith (Eds.), The Computer and Literary Studies, University Press, Edinburgh, 1973.

[33] J. Kounios, P.J. Holcomb, Concreteness effects in semantic processing—erp evidence supporting dual-coding theory, Journal of Experimental Psychology: Learning, Memory and Cognition 20 (4) (1994) 804–823.

[34] H. Kucera, W.N. Francis, Computational Analysis of Present-Day American English, Brown University Press, Providence, 1967.

[35] M. Kutas, S.A. Hillyard, Reading senseless sentences: brain potentials reflect semantic incongruity, Science 207 (1980) 203–205.

[36] C.M. MacLeod, Half a century of research on the Stroop effect: an integrated review, Psychological Bulletin 109 (1991) 163–203.

[37] A.J. Marcel, Conscious and unconscious perception: experiments on visual masking and word recognition, Cognitive Psychology 15 (1983) 197–237.

[38] P. Marí-Beffa, G. Houghton, M.A.F. Estévez, L.J. Fuentes, Word-based grouping affects the prime task effect on semantic priming, Journal of Experimental Psychology: Human Perception and Performance 26 (2) (2000a) 469–479.

[39] P. Marí-Beffa, L.J. Fuentes, A. Catena, G. Houghton, The prime task effect and negative priming: evidences for the automaticity of semantic processing from distractors, Memory and Cognition 28 (4) (2000b) 635–647.

[40] A. Martin, C.L. Wiggs, L.G. Ungerleider, J.V. Haxby, Neural correlates of category-specific knowledge, Nature 379 (6566) (1996) 649–652.

[41] M. Martín-Loeches, J.A. Hinojosa, G. Gómez-Jarabo, F.J. Rubia, The recognition potential: an ERP index of lexical access, Brain and Language 70 (1999) 364–384.

[42] M. Martín-Loeches, J.A. Hinojosa, G. Gómez-Jarabo, F.J. Rubia, An early electrophysiological sign of semantic processing in basal extrastriate areas, Psychophysiology 38 (2001a) 114–125.

[43] M. Martín-Loeches, J.A. Hinojosa, C. Fernández-Frías, F.J. Rubia, Functional differences in the semantic processing of concrete and abstract words, Neuropsychologia 39 (2001b) 1086–1096.

[44] M. Martín-Loeches, J.A. Hinojosa, P. Casado, F. Muñoz, L. Carretié, C. Fernández-Frías, M.A. Pozo, The recognition potential and repetition effects, International Journal of Psychophysiology 43 (2002) 155–166.

[45] L. Maxfield, Attention and semantic priming: a review of the prime task effects, Consciousness and Cognition 6 (1997) 204–218.

[46] J.L. McClelland, D.E. Rumelhart, An interactive activation model of context effects in letter perception: Part 1. An account of basic findings, Psychological Review 88 (1981) 315–407.

[47] C.J. Moore, C.J. Price, A functional neuroimaging study of the variables that generate category-specific object processing differences, Brain 122 (1999) 943–962.

[48] C.J. Mummery, T. Shallice, C.J. Price, Dual-process model in semantic priming: a functional imaging perspective, Neuroimage 9 (1999) 516–525.

[49] J.H. Neely, Semantic priming and retrieval from lexical memory: roles of inhibitionless spreading activation and limited-capacity attention, Journal of Experimental Psychology. General 106 (1977) 226–254.

[50] J.H. Neely, Semantic priming effects in visual word recognition: a selective review of current findings and theories, in: D. Besner, G.W. Humphreys (Eds.), Basic Processes in Reading. Visual Word Recognition, Erlbaum, N.J. Hillsdale, 1991, pp. 264–336.

[51] J.H. Neely, T.A. Kahan, Is semantic activation automatic? A critical re-evaluation, in: Henry L. Roediger III, James S. Nairne, Ian Neath, Aimée M. Surprenant (Eds.), The Nature of Remembering, American Psychological Association, Washington D.C., 2001, pp. 69–94.

[52] A.P. Rudell, J. Hua, The recognition potential and conscious awareness, Electroencephalography and Clinical Neurophysiology 98 (1996) 309–318.

[53] A.P. Rudell, J. Hua, The recognition potential, word difficulty, and individual reading ability: on using event-related potentials to study perception, Journal of Experimental Psychology. Human Perception and Performance 23 (1997) 1170–1195.

[54] M.C. Smith, L. Theodor, P.E. Franklin, The relationship between contextual facilitation and depth of processing, Journal of Experimental Psychology. Learning, Memory and Cognition 9 (1983) 697–712.

[55] J.A. Stolz, D. Besner, Role of set in visual word recognition: activation and activation blocking as nonautomatic processes, Journal of Experimental Psychology: Human Perception and Performance 22 (5) (1996) 1166–1177.

[56] L.K. Tyler, E.A. Stamatakis, E. Dick, P. Bright, P. Fletcher, H. Moss, Objects and their actions: evidence for a neurally distribute semantic system, Neuroimage 18 (2) (2003) 542–557.

[57] B. Valdés, Catena, P. Marí-Beffa, Automatic and controlled semantic processing: a masked prime-task effect. Consciousness and Cognition (in press).

[58] F. Waszak, B. Hommel, A. Allport, Task-switching and long-term priming: role of episodic stimulus-task bindings in task-shift costs, Cognitive Psychology 46 (2003) 361–413.

[59] J. Weckerly, M. Kutas, An electrophysiological analysis of animacy effects in the processing of object relative sentences, Psychophysiology 36 (5) (1999) 559–570.

[60] M.D. Wilson, The MRC psycholinguistic database: machine readable dictionary, version 2, Behavioural Research Methods, Instruments and Computers 20 (1) (1988) 6–11.